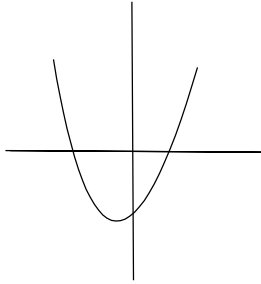


10.1 Practice - Function Notation

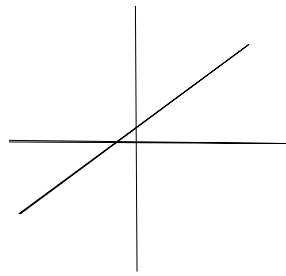
Solve.

1) Which of the following is a function?

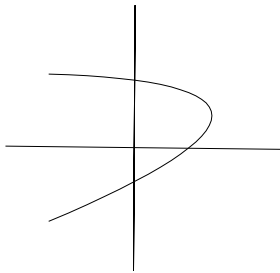
a)



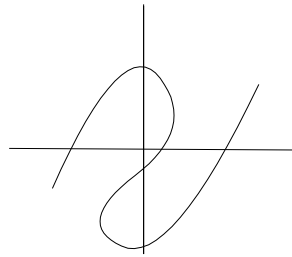
b)



c)



d)



e) $y = 3x - 7$

f) $y^2 - x^2 = 1$

g) $\sqrt{y} + x = 2$

h) $x^2 + y^2 = 1$

Specify the domain of each of the following functions.

2) $f(x) = -5x + 1$

3) $f(x) = \sqrt{5 - 4x}$

4) $s(t) = \frac{1}{t^2}$

5) $f(x) = x^2 - 3x - 4$

6) $s(t) = \frac{1}{t^2 + 1}$

7) $f(x) = \sqrt{x - 16}$

8) $f(x) = \frac{-2}{x^2 - 3x - 4}$

9) $h(x) = \frac{\sqrt{3x - 12}}{x^2 - 25}$

10) $y(x) = \frac{x}{x^2 - 25}$

Evaluate each function.

11) $g(x) = 4x - 4$; Find $g(0)$

13) $f(x) = |3x + 1| + 1$; Find $f(0)$

15) $f(n) = -2| -n - 2| + 1$; Find $f(-6)$

17) $f(t) = 3^t - 2$; Find $f(-2)$

19) $f(t) = |t + 3|$; Find $f(10)$

21) $w(n) = 4n + 3$; Find $w(2)$

23) $w(n) = 2^{n+2}$; Find $w(-2)$

25) $p(n) = -3|n|$; Find $p(7)$

27) $p(t) = -t^3 + t$; Find $p(4)$

29) $k(n) = |n - 1|$; Find $k(3)$

31) $h(x) = x^3 + 2$; Find $h(-4x)$

33) $h(x) = 3x + 2$; Find $h(-1 + x)$

35) $h(t) = 2| -3t - 1| + 2$; Find $h(n^2)$

37) $g(x) = x + 1$; Find $g(3x)$

39) $g(x) = 5^x$; Find $g(-3 - x)$

12) $g(n) = -3 \cdot 5^{-n}$; Find $g(2)$

14) $f(x) = x^2 + 4$; Find $f(-9)$

16) $f(n) = n - 3$; Find $f(10)$

18) $f(a) = 3^{a-1} - 3$; Find $f(2)$

20) $w(x) = x^2 + 4x$; Find $w(-5)$

22) $w(x) = -4x + 3$; Find $w(6)$

24) $p(x) = -|x| + 1$; Find $p(5)$

26) $k(a) = a + 3$; Find $k(-1)$

28) $k(x) = -2 \cdot 4^{2x-2}$; Find $k(2)$

30) $p(t) = -2 \cdot 4^{2t+1} + 1$; Find $p(-2)$

32) $h(n) = 4n + 2$; Find $h(n + 2)$

34) $h(a) = -3 \cdot 2^{a+3}$; Find $h(\frac{a}{4})$

36) $h(x) = x^2 + 1$; Find $h(\frac{x}{4})$

38) $h(t) = t^2 + t$; Find $h(t^2)$

40) $h(n) = 5^{n-1} + 1$; Find $h(\frac{n}{2})$